Ideal-Gas Processes and Cycle Energy Accounting

MECHANICAL-ENGINEERING · mechanical-engineering-mechanical-engineering-professional-subjects · BoardPrep lesson v1

Official examination scope

The PRC August 2026 ME program lists Power Plant Engineering (35%), Mathematics (35%), and Machine Design, Materials and Shop Practice (30%); it gives no thermal-science subtopic weight. This is foundational thermodynamics instruction from existing BoardPrep material. Verify detailed scope in current PRC syllabus resources; no frequency claim is made.

Learning objectives

- Use an ideal-gas relation for a fixed-mass, fixed-volume process with absolute temperature.
- Compute work for a stated isothermal ideal-gas expansion.
- Calculate cycle net work and thermal efficiency from heat input and rejection.

1. State the gas model

For fixed mass of ideal gas, pV = mRT. At fixed volume and mass, p/T is constant, so p2 = p1(T2/T1). Use kelvins. The relation does not apply unchanged when mass or volume changes.

2. Calculate isothermal work

For quasi-equilibrium isothermal ideal-gas expansion, work by the gas is W = mRT ln(V2/V1). With R in kJ/(kg K), mass in kg, and temperature in K, work is kJ. Expansion is positive work by the gas under this convention; compression reverses the sign.

3. Account for a cycle

For a heat-engine cycle, net stored energy change is zero. With heat input Qin and rejected heat Qout, Wnet = Qin - Qout and thermal efficiency eta = Wnet/Qin. Efficiency is dimensionless; multiply by 100 only when reporting a percent.

4. Check units and balances

Write constraints and sign convention, convert units, and check expected signs. Engine work should be less than heat input, and efficiency should be dimensionless. A number without its process assumptions is not reproducible.

Fixed-volume pressure change

Question: An ideal gas at 150 kPa and 300 K is heated to 450 K at constant mass and volume.
1. Use p2/p1 = T2/T1.
2. p2 = 150(450/300) kPa.
3. The ratio is 1.5, so p2 = 225 kPa.
Answer: Final pressure = 225 kPa.
Check: A 50% rise in absolute temperature gives a 50% pressure rise at fixed volume.

Heat-engine work and efficiency

Question: A cycle receives 900 kJ and rejects 540 kJ per cycle.
1. Wnet = 900 - 540 = 360 kJ.
2. eta = 360/900 = 0.40.
3. If requested, 0.40 x 100% = 40%.
Answer: Net work is 360 kJ per cycle and efficiency is 0.40 (40%).
Check: Output work is less than heat input and efficiency has no unit.

Common misconceptions

- Using Celsius instead of kelvins in a pressure-temperature ratio.
- Using p/T = constant when volume or gas mass is not fixed.

- Treating isothermal ideal-gas expansion work as a rectangular p-V area.
- Dividing rejected heat by input heat instead of using net work for engine efficiency.

Practice patterns

- Find a state change from a stated ideal-gas process constraint.
- Calculate expansion work from mass, R, temperature, and volume ratio.
- Determine cycle work and efficiency from input and rejected heat.

Official scope references

Official scope: PRC Mechanical Engineering Syllabus Resources — Professional Regulation Commission; Professional Regulatory Board of Mechanical Engineering — https://www.prc.gov.ph/Pages/PRBv4/MechanicalEngineering.htm
Official scope: PRC Program of the August 2026 Mechanical Engineers Licensure Examination — Professional Regulation Commission; Professional Regulatory Board of Mechanical Engineering — https://www.prc.gov.ph/sites/default/files/exam%20program%20aug%202026%20mele%20signed.pdf

Instructional references

Instructional reference: Introduction to Engineering Thermodynamics — Claire Yu Yan; BCcampus Open Education — https://pressbooks.bccampus.ca/thermo1/
Instructional reference: University Physics Volume 2 — OpenStax, Rice University — https://openstax.org/details/books/university-physics-volume-2

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No claim about question frequency or examination weighting is made.